

Embedded System

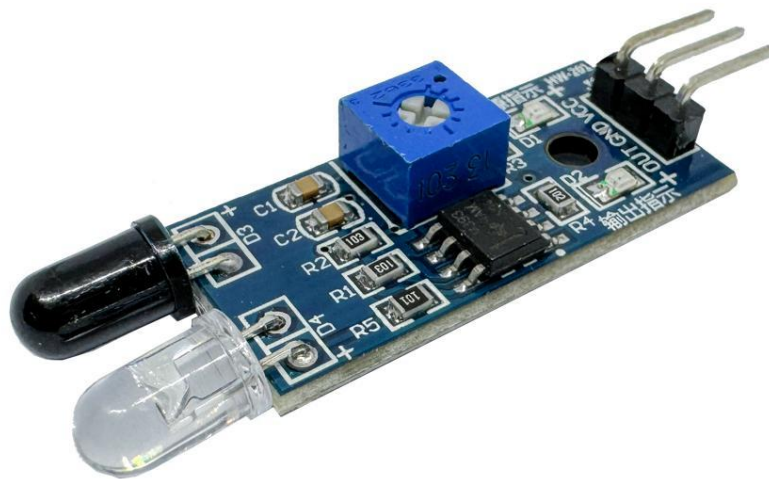
1) Obstacle Detection using Infrared (IR) Sensor

Experiment Description:

An obstacle detection system was developed using an IR sensor interfaced with an Arduino board to identify the presence of nearby objects.

Explanation:

The IR sensor emits infrared radiation and detects the reflected signal from an object. Based on the reflection, the sensor provides a digital output to the Arduino. The Arduino processes this signal to indicate object detection through an LED or serial output.



Code:

```
int irPin = 2;
```

```
int led = 13;
```

```
void setup() {
```

```
    pinMode(irPin, INPUT);
```

```
    pinMode(led, OUTPUT);
```

```
    Serial.begin(9600);
```

```
}
```

```
void loop() {
```

```
    int val = digitalRead(irPin);
```

```
    if (val == LOW) {
```

```
        digitalWrite(led, HIGH);
```

```
        Serial.println("Object Detected");
```

```
    } else {
```

```
        digitalWrite(led, LOW);
```

```
        Serial.println("No Object");
```

```
    }
```

```
}
```

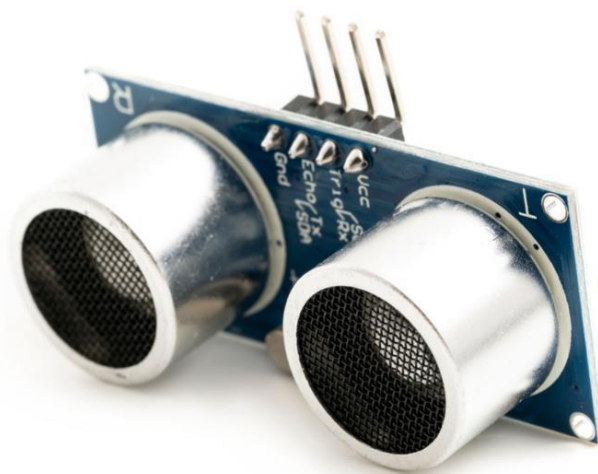
2) Distance Measurement using Ultrasonic Sensor

Experiment Description:

A distance measurement system was implemented using an ultrasonic sensor interfaced with Arduino to calculate the distance of an object.

Explanation:

The ultrasonic sensor transmits sound waves and receives the reflected echo signal. The time taken for the echo to return is measured by the Arduino. This time is converted into distance using the speed of sound.



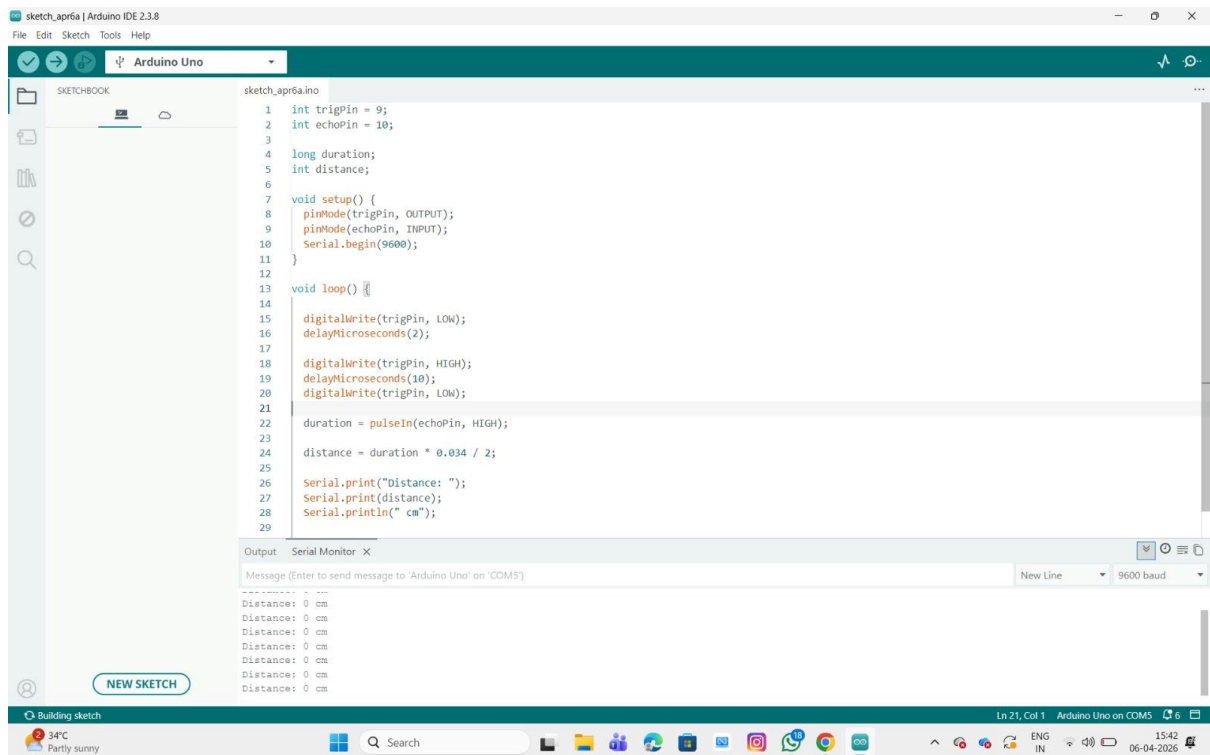
Code:

```
int trigPin = 9;  
int echoPin = 10;  
long duration;
```

```
int distance;
```

```
void setup() {  
  pinMode(trigPin, OUTPUT);  
  pinMode(echoPin, INPUT);  
  Serial.begin(9600);  
}
```

```
void loop() {  
  digitalWrite(trigPin, LOW);  
  delayMicroseconds(2);  
  
  digitalWrite(trigPin, HIGH);  
  delayMicroseconds(10);  
  digitalWrite(trigPin, LOW);  
  
  duration = pulseIn(echoPin, HIGH);  
  distance = duration * 0.034 / 2;  
  
  Serial.print("Distance: ");  
  Serial.println(distance);  
  
  delay(500);  
}
```



3) Temperature and Humidity Measurement using DHT Sensor:

Experiment Description:

A temperature and humidity monitoring system was developed using a DHT11 sensor interfaced with Arduino, utilizing an external library for data acquisition.

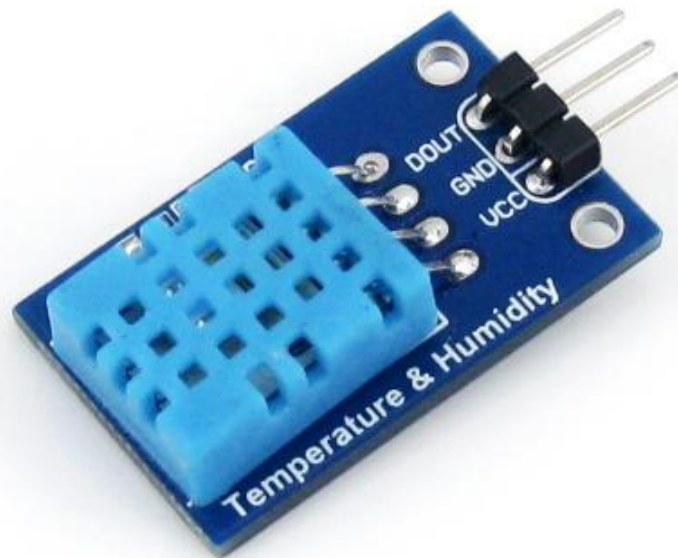
Explanation:

The DHT11 sensor measures ambient temperature and humidity using internal sensing elements. A dedicated library is installed to enable communication between the sensor and Arduino. The Arduino retrieves the data and displays the measured values on the serial monitor.

Library Used:

- DHT Sensor Library (Adafruit)

- Adafruit Unified Sensor Library



Code:

```
#include <DHT.h>
```

```
#define DHTPIN 2
```

```
#define DHTTYPE DHT11
```

```
DHT dht(DHTPIN, DHTTYPE);
```

```
void setup() {
```

```
  Serial.begin(9600);
```

```
dht.begin();  
  
}  
  
void loop() {  
    float temp = dht.readTemperature();  
    float hum = dht.readHumidity();  
  
    if (isnan(temp) || isnan(hum)) {  
        Serial.println("Sensor Error");  
        return;  
    }  
  
    Serial.print("Temperature: ");  
    Serial.print(temp);  
    Serial.print(" °C ");  
  
    Serial.print("Humidity: ");  
    Serial.print(hum);  
    Serial.println(" %");  
  
    delay(2000);  
}
```

sketch_apr6a | Arduino IDE 2.3.8

File Edit Sketch Tools Help

Arduino Uno

LIBRARY MANAGER

DHT

Type: All

Topic: All

EduIntro by Arduino LLC

Library used for super-fast introduction workshops is intended to be used with Arduino UNO / ...

More info

0.0.16 **INSTALL**

AM2302-Sensor by Frank Häfele <mail@frankhaefele.de>

This library read temperature and humidity from the AM2302 (aka DHT22) sensor. The AM2302 sensor...

More info

1.5.0 **INSTALL**

AM2302-Sensor-Pi-Pico by Frank Häfele...

This library read temperature and humidity from the AM2302 (aka DHT22) sensor on Pi Pico. The...

More info

1.0.0 **INSTALL**

AM232X by Rob Tillaart <rob.tillaart@gmail.com>

Arduino library for AM2320 AM2321 and AM2323 I2C temperature and humidity sensor. Supports AM232...

sketch_apr6a.ino

```
1 #include <DHT.h>
2
3 #define DHTPIN 2
4 #define DHTTYPE DHT11
5
6 DHT dht(DHTPIN, DHTTYPE);
7
8 void setup() {
9   Serial.begin(9600);
10  dht.begin();
11 }
12
13 void loop() {
14
15   float humidity = dht.readHumidity();
16   float temperature = dht.readTemperature();
17
18   if (isnan(humidity) || isnan(temperature)) {
19     Serial.println("Sensor error");
20     return;
21   }
22
23   Serial.print("Temperature: ");
24   Serial.print(temperature);
25   Serial.print("C");
26   Serial.print(" Humidity: ");
27   Serial.print(humidity);
28   Serial.print("%\n");
29 }
```

Output Serial Monitor X

Message (Enter to send message to 'Arduino Uno' on 'COM5')

New Line 9600 baud

Temperature: 33.00 °C
Humidity: 35.00 %

Temperature: 33.30 °C
Humidity: 35.00 %

Temperature: 33.40 °C
Humidity: 35.00 %

Temperature: 33.50 °C
Humidity: 36.00 %

Building sketch

34°C Partly sunny

Ln 4, Col 22 Arduino Uno on COM5 6

15:37 06-04-2026